

The examination is being carried out on the **following application documents**

Description, Pages

1-175 as originally filed

Claims, Numbers

1-4 filed in electronic form on 18-01-2019

Drawings, Sheets

1/8-8/8 as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

- D1 WO 2008/039218 A2 (PHARMACYCLICS INC [US]; HONIGBERG LEE [US]; VERNER ERIK [US]; PAN ZHEN) 3 April 2008 (2008-04-03)
- D2 WO 2007/136790 A2 (MANNKIND CORP [US]; FLYNN GARY A [US]; LEE SANDRA AEYOUNG [US]; FARIS) 29 November 2007 (2007-11-29)

The following document is cited by the Examiner. A copy of the document is annexed to the communication and the numbering will be adhered to in the rest of the procedure.

- D3 Anna Smith: "Imbruvica fails Phase III pancreatic cancer trial - PharmaTimes",
, 1 January 2019 (2019-01-01), pages 1-2, XP055593391,
Retrieved from the Internet:
URL:http://www.pharmatimes.com/news/imbruvica_fails_phase_iii_pancreatic_cancer_trial_1275758
[retrieved on 2019-06-03]

D3 (Smith et al. 2019)

teaches that ibrutinib failed to treat metastatic pancreatic cancer in combination therapy.

Art. 83 EPC

D3 teaches that ibrutinib failed to treat metastatic pancreatic cancer. In view of this teaching and in view of the fact that the present application does not provide any data demonstrating the effective treatment of solid cancers, in particular pancreatic cancers, it is not plausible without further evidence that ibrutinib treats solid cancers over the entire scope of the claims and it would require a research program to identify the specific solid cancer which is susceptible to ibrutinib treatment *in vivo*.

Art. 54 EPC

The ED maintains the view that the subject-matter of claims 1, 2, 4 lacks novelty in view of D1.

D1 (WO2008039218) claims, in particular claim 8; §0041; compounds of §0205; §00412, table 3

discloses compounds of present formula I, preferably compounds of claim 8 of D1, as inhibitors of Btk for use in the treatment of cancer, preferably a B-cell proliferative disorder such as lymphoma (§0034), e.g. Burkitt's lymphoma (= a solid tumor). It is explicitly disclosed/suggested to **treat pancreatic cancer and other solid** or hematological tumors (§0041). Compound 4, i.e. the racemate of ibrutinib is the preferred compound of the examples, see examples 3-5 (§00412, table 3). Ibrutinib (i.e. the **R**-enantiomer) thereof is disclosed as compound 13 (example 1B + table 3). Ibrutinib (i.e. the **R enantiomer**) was furthermore demonstrated to be **more active** than its racemate compound 4 on Btk (table 3), compare compound 13 with 4. Thus, ibrutinib (racemate) and even more ibrutinib (R) enantiomer are particularly highlighted. It is concluded that no selection is required to arrive from D1 at the specific combination of ibrutinib for the treatment of solid tumors [disclosed in [0041]], in particular pancreatic tumors [disclosed in [0041] or solid lymphomas (such as Burkitt's lymphoma [0034]).

In contrast thereto, D1 does not disclose the treatment of sarcoma or carcinoma.

Art. 56 EPC

If however, the applicant considers that the skilled person would extrapolated the efficacy of ibrutinib from one solid tumor to another one despite of the negative teaching of D3, the following applies.

D1 may be considered to be the closest prior art as it discloses ibrutinib for use in the treatment of solid tumors, e.g. Burkitt's lymphoma (§0029) or pancreatic cancer (§0041).

Present claims - in as far as novel - differ therefrom in that they relate to the presently claimed solid tumors.

Thus, the problem to be solved in view of D1 is to find an alternative tumor which might be susceptible to the treatment with ibrutinib.

The ED notices that the present claims are not restricted to breast cancer (for which the applicant provided STIN for the parent application), but relates to a variety of further solid tumors. No data have been provided for any of the claimed tumors. Thus, the applicant extrapolated from the anti-tumor activity evidenced in the example to the general usefulness of the compounds in treating the claimed solid tumors.

Similarly, D1 explicitly teaches the usefulness of ibrutinib in the treatment of solid tumors, such as Burkitt's lymphoma (= a solid tumor), pancreatic cancer and other solid or hematological tumors (§0041).

Thus, by applying the same level of skill, the skilled person would extrapolate from the anti-tumor activity against solid tumors of the claimed compounds as suggested in D1 to the usefulness of said compounds in the treatment of the presently claimed tumors.

Thus, the subject-matter of the claims is obvious in view of D1.

The applicant's argument that the D1 would not suggest the specific claimed solid tumor types does not seem to be decisive here. It appears more important to the ED whether or not the skilled person would extrapolate from the usefulness of ibrutinib from one solid tumor type (for which no data were provided in the application) to another solid tumor type (for which no data were provided in the application). The later provided data suggesting a potential usefulness of ibrutinib in certain mammary cancers support the initial teaching of the application that solid mammary cancers may be treated by ibrutinib, however, they also support the initial teaching of D1 that ibrutinib indeed treats solid tumors.

To overcome the objection it appears necessary to provide data supporting the potential use of ibrutinib for each of the claimed cancer types.